PULSE AI

Technical Documentation

Shenzhen International FinTech Competition 2026 (FinTechathon), International Track, Topic E

Repository: github.com/EyaAzzabi/Pulse (branch main)

1. Overview

PULSE AI is a cash-flow copilot for small and medium enterprises. It turns transactions, invoices and recurring obligations into a deterministic forecast, detects upcoming shortfalls, simulates rescue strategies in a Financial Digital Twin and lets AI agents propose actions that only run after a human with the right role approves them.

Every figure shown to the user comes from the deterministic engines. The language model chooses tools and phrases answers; it never invents numbers.

2. Problem

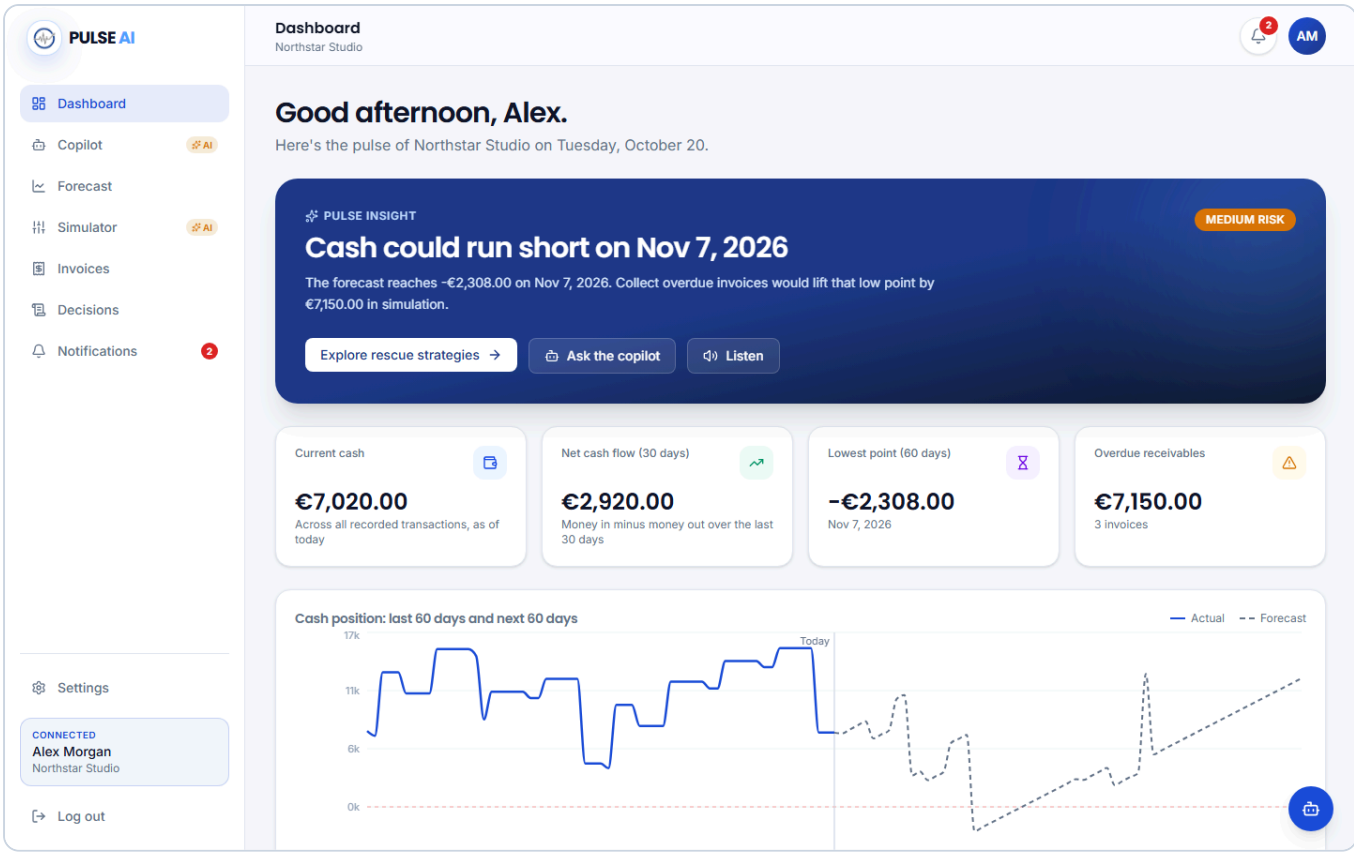
SMEs often see a cash crisis only when it arrives: late customer payments, clustered supplier bills and tax dates combine into a negative balance weeks later. Spreadsheets do not show the date of the shortfall or which action helps most.

3. Solution

- **Onboarding** with email verification code, company profile, role, and data connection (Excel/CSV upload with column mapping, Google Sheets link, Northstar Studio sample company, or quick estimates).
- **Dashboard**: current cash, 30-day net flow, lowest point in 60 days, overdue receivables, insight card, upcoming 30 days, overdue aging, what changed.
- **Forecast**: day-by-day cash projection with known and uncertain components.
- **Simulator**: what-if scenarios (customer delay, supplier delay, revenue/expense change, financing, injections).
- **Copilot**: multi-agent chat with conversation history, voice input and read-aloud, and a visible agent trace.
- **Decision Center**: propose, approve, execute, with audit log.
- **Notifications** and **Admin console** (users, organizations, account review, service status, audit).

4. System architecture

```
Browser (React 19 + Vite + Tailwind)            n8n (optional, signed webhooks)
  Landing / Onboarding / Dashboard / Copilot       verification code, invoice reminder,
  Forecast / Simulator / Decisions / Admin         cash risk alert, digest, recovery plan
             | HTTPS + JWT bearer                          ^  HMAC/header secret
             v                                            |
FastAPI backend  ---------------------------------------+
  api/        auth, onboarding, imports, financial, forecast, simulations,
              recommendations, actions, copilot, notifications, admin
  agents/     orchestrator (OpenAI tool calling, deterministic fallback) + tools.py
  forecasting/ deterministic cash forecast engine (Decimal)
  simulations/ Financial Digital Twin (scenario recalculation)
  services/   rescue plans, alerts, notifications, OTP, email, tabular import
             | SQLAlchemy 2 + Alembic
             v
Supabase Postgres (session pooler)   |  SQLite for the automated test suite
```



Dashboard, Northstar Studio, captured from the running app

5. Core scenario flow

```
Financial data -> Summary + risk -> 90-day forecast -> shortfall detected
  -> Digital Twin scenarios -> ranked rescue plans -> agent proposes action
  -> PENDING_APPROVAL -> human approves (role checked) -> execute
  -> QUEUED (no webhook) or dispatched via n8n -> audit log + notification
```

6. Data model

Entity	Purpose
User, Organization, Membership	Tenancy and role per organization (OWNER, FINANCE_MANAGER, ACCOUNTANT, VIEWER); is_admin, account_status, email_verified
Transaction, Invoice, RecurringObligation	Financial inputs; money stored as Decimal
ForecastRun / points	Stored forecasts with starting cash and risk
Scenario / SimulationResult	Digital Twin runs and assumptions
Action, Recommendation	Proposed actions with status PENDING_APPROVAL, APPROVED, QUEUED, SENT, FAILED
Conversation, Message	Copilot memory with stored agent trace
Notification, AuditLog, AccountReview, EmailOtp	Alerts, traceability, admin review, hashed verification codes

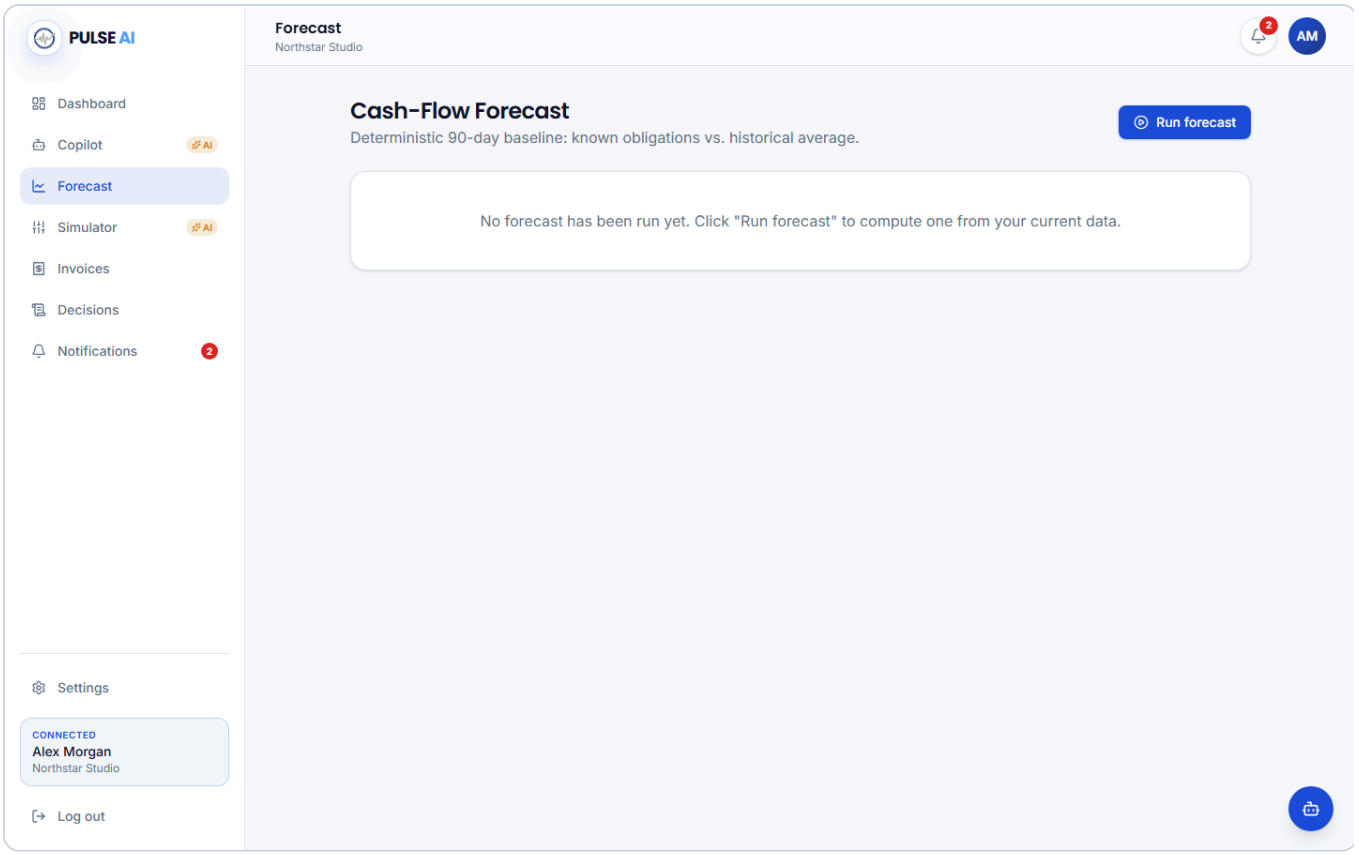
7. Core algorithms

7.1 Current liquidity and summary

Current cash = opening balance + sum of all recorded transactions up to today. Runway, 30-day net flow and overdue receivables are computed from the same inputs.

7.2 Deterministic forecast

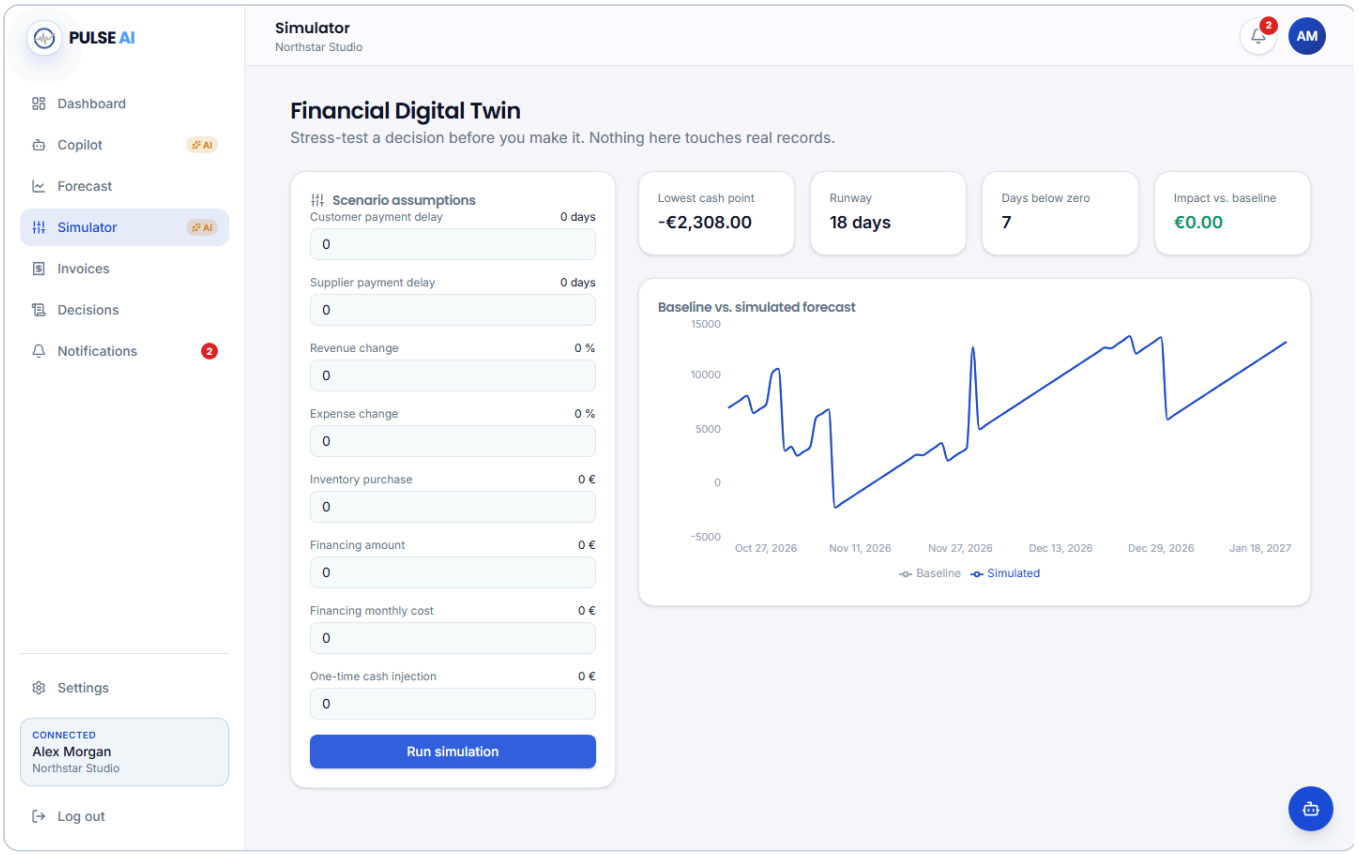
Starting from current liquidity, each future day adds *known* flows (open invoices on their due date, recurring obligations on their schedule) and an *uncertain* baseline derived from the historical daily average of categories not already modeled as recurring (to avoid double counting). Output: daily balance, lowest point, first shortfall date, risk level and reasons.



Forecast page

7.3 Financial Digital Twin

The simulator re-runs the same engine with modified inputs: shifting customer invoice dates (optionally for one named customer), shifting supplier payments, scaling revenue or expenses, adding a purchase, financing with monthly cost, collecting overdue invoices within N days. It returns baseline vs scenario curves and the change in the lowest point.



Simulator with a scenario result

7.4 Rescue plans

Candidate strategies (collect overdue invoices, negotiate supplier extensions, bridge with a short-term credit line, labeled illustrative) are each simulated and ranked by the improvement of the lowest cash point. On Northstar Studio the measured improvements are +EUR 7,150, +EUR 4,560 and +EUR 4,372.

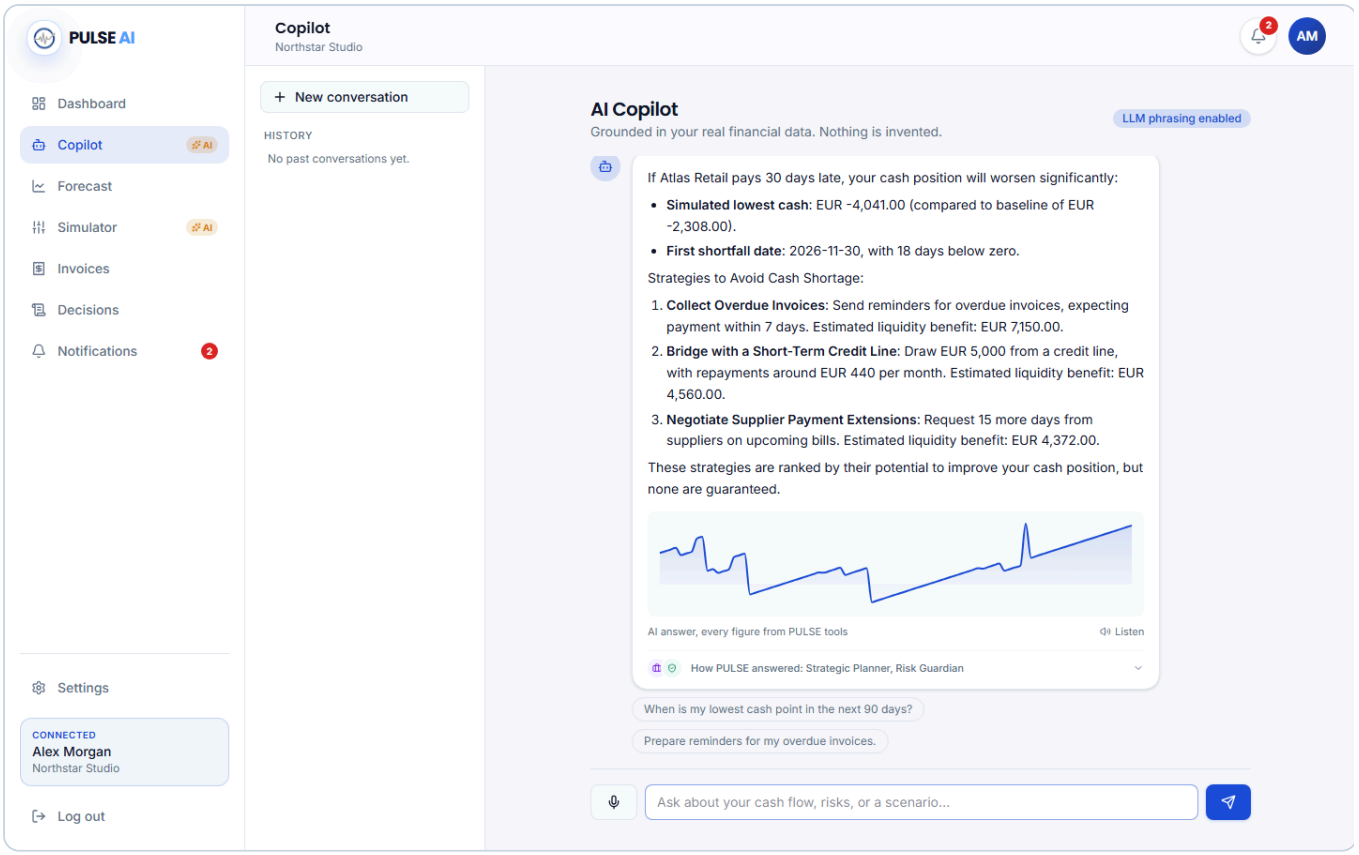
7.5 Alerts

Summary and forecast requests evaluate rules (projected shortfall, overdue invoices) and create notifications with a 7-day deduplication window.

8. Agentic AI architecture

Agent	Tools	Role
Financial Analyst	get_financial_summary, list_invoices	Current position
Forecast Analyst	forecast_cash	Future balance and shortfall
Strategic Planner	simulate_scenario, get_rescue_plans	What-if and strategy ranking
Workflow Coordinator	propose_invoice_reminder	Creates PENDING_APPROVAL actions only
Risk Guardian	role checks, final review	Denies tools above the user role

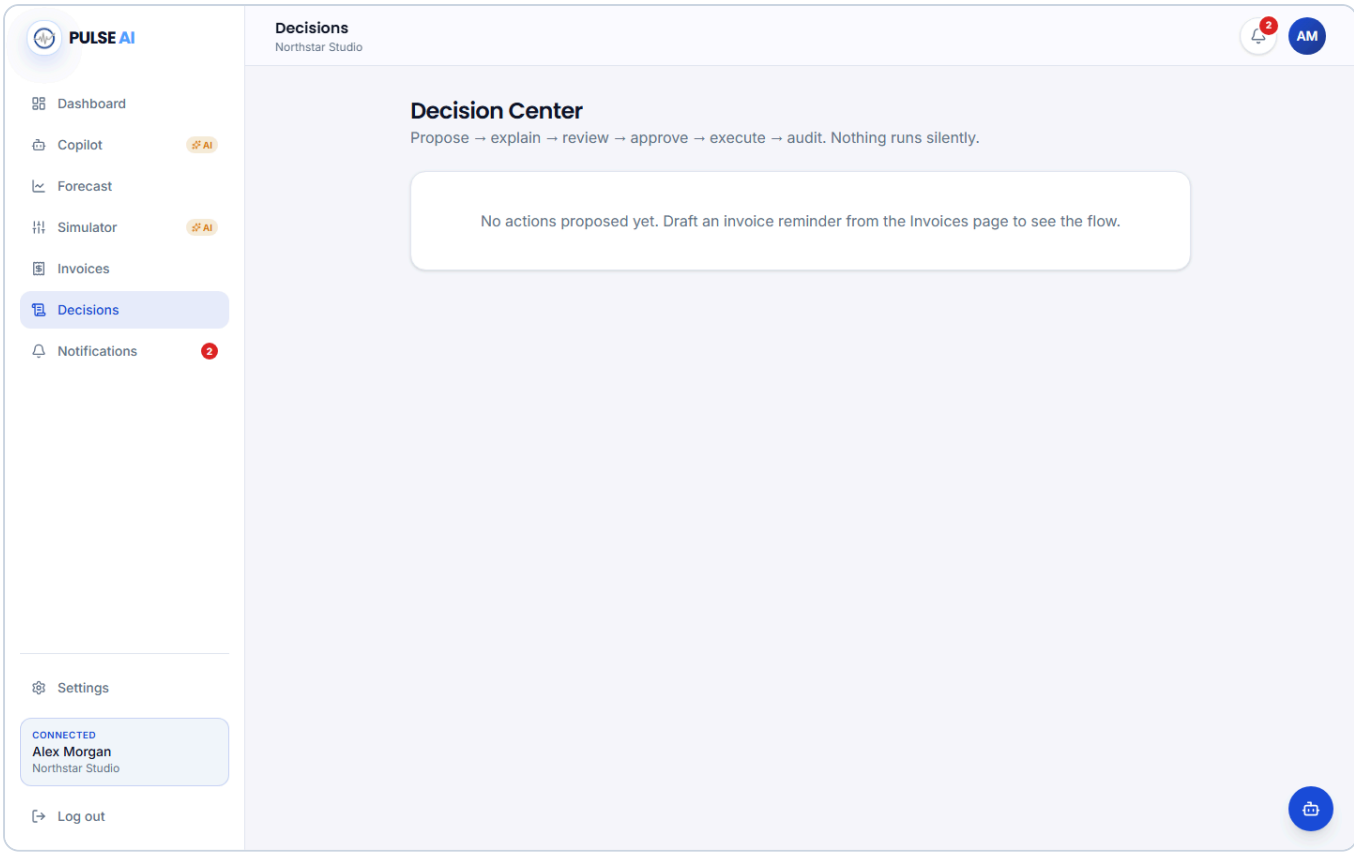
The orchestrator uses OpenAI tool calling (gpt-4o-mini) with at most 6 steps. Tool results are deterministic engine outputs. If the model is unavailable, a deterministic intent router answers from the same tools. The trace (agents and tools used) is returned and displayed under each answer.



Copilot answer for a late-payment question, with agent trace (Strategic Planner, Risk Guardian)

9. Human-in-the-loop execution

Actions follow PROPOSE (ACCOUNTANT or above) then APPROVE (role checked) then EXECUTE. Executing an unapproved action returns HTTP 409; executing twice returns 409 (idempotent). If no n8n webhook is configured, the action is recorded as QUEUED with an explicit message and is never reported as SENT.



Decision Center

10. Security design

JWT authentication, bcrypt password hashes, email verification codes stored as HMAC hashes with 10-minute expiry, 5 attempts and 60 s resend cooldown, organization-scoped queries, role hierarchy, admin-only console, audit log, server-side Google Sheets URL reconstruction (SSRF guard), secrets only in environment variables. Full detail and known risks are in PULSE_Security_Self_Assessment.pdf.

11. Integrations

Supabase Postgres (session pooler), OpenAI API, SMTP (Gmail app password) for verification codes, n8n workflows in `n8n/workflows/` (verification code, invoice reminder, cash risk alert, financial digest, approved recovery plan) protected by a header secret.

12. Testing

87 automated backend tests (pytest, SQLite) pass: auth and OTP, onboarding, imports, financial summary, forecast and simulation regressions, rescue plans, agents and tool permissions, actions workflow, notifications, admin. Output in `execution_evidence/test_output.txt`. Frontend: TypeScript strict build passes.

13. Deployment

```
cd backend && python -m venv .venv && .venv/Scripts/activate
pip install -r requirements.txt
cp .env.example .env      # DATABASE_URL, JWT secret, optional OPENAI_API_KEY, SMTP_*
alembic upgrade head
python scripts/seed_demo.py         # Northstar Studio company
uvicorn app.main:app --reload       # http://127.0.0.1:8000/docs
cd ../frontend && npm install && npm run dev
cd backend && pytest -q
```

14. Limitations and roadmap

- Forecast is deterministic and rule-based; no ML model is trained yet.
- External sending depends on n8n/SMTP configuration; otherwise actions stay QUEUED.
- No direct bank or ERP connector yet (Odoo connector planned); data comes from file, Google Sheets or manual entry.
- No rate limiting, MFA, or field-level encryption yet (see security report).
- Credit line strategy uses illustrative terms, not a lender offer.